

EXPRESS.JS ROADMAP

Phase 1: Backend Fundamentals

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└─ What is Backend?

└─ Client vs Server

└─ HTTP Protocol

└─ Request & Response

└─ HTTP Methods (GET, POST, PUT, PATCH, DELETE)

└─ Status Codes

└─ REST API Basics

└─ JSON



Phase 2: Node.js Fundamentals

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└─ Node.js Introduction

└─ npm

└─ package.json

└─ Modules

└─ require vs import

└─ File System (fs)

- └─ Path Module
- └─ OS Module
- └─ Environment Variables
- └─ Nodemon



Phase 3: Express.js Basics

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- └─ Installing Express
- └─ First Express Server
- └─ app.listen()
- └─ app.get()
- └─ app.post()
- └─ app.put()
- └─ app.patch()
- └─ app.delete()
- └─ Sending Responses
- └─ JSON Response
- └─ HTML Response
- └─ Route Parameters



Phase 4: Routing

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└─ Express Router

└─ Route Parameters

└─ Query Parameters

└─ Nested Routes

└─ Route Organization

└─ Modular Routing



Phase 5: Middleware

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└─ What is Middleware?

└─ next()

└─ Built-in Middleware

└─ express.json()

└─ express.urlencoded()

└─ Static Files

└─ Custom Middleware

└─ Logger Middleware

└─ Authentication Middleware

|— Authorization Middleware

|— Global Middleware



Phase 6: MVC Architecture

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|— What is MVC?

|— Models

|— Views (Optional)

|— Controllers

|— Routes

|— Services

|— Utilities

|— Config Folder

|— Clean Project Structure



Phase 7: REST API Development

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|— CRUD Operations

|— API Design

└─ REST Naming Convention

└─ Versioning

└─ Response Structure

└─ Pagination

└─ Filtering

└─ Sorting

└─ Searching

└─ API Testing (Postman)



Phase 8: Database Integration

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└─ MongoDB Basics

└─ Mongoose

└─ Schema

└─ Models

└─ CRUD

└─ Validation

└─ Populate

└─ Relationships

└─ Indexes

└─ Aggregation Basics



Phase 9: Validation

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└─ Input Validation

└─ express-validator

└─ Joi

└─ Required Fields

└─ Custom Validation

└─ Error Messages

└─ Sanitization



Phase 10: Authentication

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└─ Password Hashing

└─ bcrypt

└─ JWT

└─ Login

└─ Register

└─ Protected Routes

- └─ Refresh Token
- └─ Email Verification
- └─ Forgot Password
- └─ Reset Password



Phase 11: Authorization

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- └─ Role Based Access
- └─ Admin
- └─ User
- └─ Permissions
- └─ Resource Ownership
- └─ Route Protection



Phase 12: Cookies & Sessions

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- └─ Cookies
- └─ cookie-parser
- └─ express-session

|— Session Storage

|— Secure Cookies

|— HttpOnly Cookies



Phase 13: Security

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|— Helmet

|— CORS

|— XSS Protection

|— SQL Injection

|— NoSQL Injection

|— Rate Limiter

|— CSRF

|— Environment Variables

|— Secure Headers



Phase 14: Logging & Performance

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- └─ Compression
- └─ Caching Basics
- └─ Error Logging
- └─ Performance Optimization
- └─ Debugging



Phase 15: Error Handling

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- └─ Try Catch
- └─ Async Errors
- └─ Global Error Handler
- └─ Custom Error Class
- └─ HTTP Errors
- └─ Validation Errors



Phase 16: File Upload

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- └─ Multer
- └─ Single Upload

|— Multiple Upload

|— Image Validation

|— Cloudinary

└ File Storage



Phase 17: Advanced Express

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|— Async/Await

|— Async Middleware

|— Event Loop Basics

|— Streams

|— Buffer

|— Cluster

|— Worker Threads

└ Performance Tips



Phase 18: Testing

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|— Jest

- └─ Supertest
- └─ Unit Testing
- └─ API Testing
- └─ Integration Testing
- └─ Mocking



Phase 19: Deployment

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- └─ Environment Variables
- └─ Render
- └─ Railway
- └─ VPS Basics
- └─ PM2
- └─ Reverse Proxy
- └─ Nginx Basics
- └─ Production Build



Phase 20: Industry-Level Project

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- └─ Authentication
- └─ CRUD
- └─ Image Upload
- └─ Search
- └─ Pagination
- └─ Role Based Access
- └─ Admin Dashboard
- └─ Email Service
- └─ Notifications
- └─ Payment Gateway
- └─ API Documentation (Swagger)
- └─ Production Deployment